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Chapter 1

Integration Bee Training — Part 1: Basics

This section focuses on mastering the foundational rules of integration, including power rules, algebraic expansions, symmetry shortcuts for definite integrals, and careful handling of absolute values. Mastery of these fundamentals is necessary before advancing to competitive speed integration techniques.

1.1 Algebraic Manipulations and Power Rule

Example 1

$$\int (2x^2 + 8x + x + 4) dx$$

Combine like terms and apply the power rule:

$$\int (2x^2 + 9x + 4) dx = \frac{2x^3}{3} + \frac{9x^2}{2} + 4x + C$$

Example 2

$$\int x^2 (x^{1/2} + x^{-1/2} + x^{-3/2}) dx$$

Distribute the x^2 term by adding the exponents:

$$\int (x^{5/2} + x^{3/2} + x^{1/2}) dx = \frac{2}{7}x^{7/2} + \frac{2}{5}x^{5/2} + \frac{2}{3}x^{3/2} + C$$

Example 3

$$\int \frac{x + \sqrt{x} + x^2}{x} dx$$

Divide each term in the numerator by the denominator:

$$\int (1 + x^{-1/2} + x) dx = x + 2\sqrt{x} + \frac{1}{2}x^2 + C$$

Example 4

$$\int \frac{x^4 + 2x^2 + 2}{x^2 + 1} dx$$

By polynomial long division or structural rearrangement:

$$\int \left(x^2 + 1 + \frac{1}{x^2 + 1} \right) dx = \frac{x^3}{3} + x + \arctan x + C$$

Example 5

$$\int \frac{(1+x)^2}{x^2} dx$$

Expand the numerator and split the fraction:

$$\int \frac{x^2 + 2x + 1}{x^2} dx = \int \left(1 + \frac{2}{x} + \frac{1}{x^2} \right) dx = x + 2 \ln |x| - \frac{1}{x} + C$$

1.2 Standard Core Primitives**Example 6**

$$\int (\sec^2 x + \sec x \tan x) dx = \tan x + \sec x + C$$

Example 7

$$\int 2^x (1 + 5^x) dx$$

Distribute the 2^x into the parentheses. Note that $2^x \cdot 5^x = (2 \cdot 5)^x = 10^x$:

$$\int (2^x + 10^x) dx = \frac{2^x}{\ln 2} + \frac{10^x}{\ln 10} + C$$

Example 8

$$\int (e^x + \sin x) dx = e^x - \cos x + C$$

Example 9

$$\int (\csc^2 x + \csc x \cot x) dx = -\cot x - \csc x + C$$

1.3 Definite Integrals and Boundary Properties**Example 10**

$$\int_{-1}^1 (1-x)^2 x dx$$

Expand the integrand:

$$\int_{-1}^1 (x^3 - 2x^2 + x) dx$$

Recognize that over a symmetric interval $[-a, a]$, all odd functions (like x^3 and x) evaluate to 0. We only need to integrate the even function component:

$$\int_{-1}^1 -2x^2 dx = 2 \int_0^1 -2x^2 dx = 2 \left[-\frac{2x^3}{3} \right]_0^1 = -\frac{4}{3}$$

Example 11

$$\begin{aligned} \int_0^1 x(\sqrt[3]{x} + \sqrt[4]{x}) dx \\ \int_0^1 (x^{4/3} + x^{5/4}) dx = \left[\frac{3}{7}x^{7/3} + \frac{4}{9}x^{9/4} \right]_0^1 = \frac{3}{7} + \frac{4}{9} = \frac{55}{63} \end{aligned}$$

Example 12

$$\int_1^2 \left(\frac{x}{2} - \frac{2}{x} \right) dx = \left[\frac{x^2}{4} - 2 \ln |x| \right]_1^2 = (1 - 2 \ln 2) - \left(\frac{1}{4} - 0 \right) = \frac{3}{4} - 2 \ln 2$$

Example 13

$$\int_0^{\pi/4} \sec x \tan x dx = \left[\sec x \right]_0^{\pi/4} = \sqrt{2} - 1$$

Example 14

$$\int_0^{\pi/4} \frac{1 + \cos^3 x}{\cos^2 x} dx = \int_0^{\pi/4} (\sec^2 x + \cos x) dx = \left[\tan x + \sin x \right]_0^{\pi/4} = 1 + \frac{\sqrt{2}}{2}$$

Example 15

$$\int_0^{\sqrt{3}/2} \frac{1}{\sqrt{1-x^2}} dx = \left[\arcsin x \right]_0^{\sqrt{3}/2} = \frac{\pi}{3}$$

Example 16

$$\int_0^{1/\sqrt{3}} \frac{x^2 - 1}{x^4 - 1} dx$$

Factor the denominator as $(x^2 - 1)(x^2 + 1)$ and cancel the common term:

$$\int_0^{1/\sqrt{3}} \frac{1}{x^2 + 1} dx = \left[\arctan x \right]_0^{1/\sqrt{3}} = \frac{\pi}{6}$$

1.4 Absolute Value Integration Strategies

Example 17

$$\int_0^2 |2x - 1| dx$$

Find the root ($x = 1/2$) and split the integral bounds to remove the absolute value:

$$\int_0^{1/2} (1 - 2x) dx + \int_{1/2}^2 (2x - 1) dx = \left[x - x^2 \right]_0^{1/2} + \left[x^2 - x \right]_{1/2}^2 = \frac{1}{4} + \left(2 - \left(-\frac{1}{4} \right) \right) = \frac{5}{2}$$

Example 18

$$\int_{-1}^1 (|x| - 2x) dx$$

Split at the root ($x = 0$):

$$\begin{aligned} \int_{-1}^0 (-x - 2x) dx + \int_0^1 (x - 2x) dx &= \int_{-1}^0 -3x dx + \int_0^1 -x dx \\ &= \left[-\frac{3}{2}x^2 \right]_{-1}^0 + \left[-\frac{1}{2}x^2 \right]_0^1 = \left(0 - \left(-\frac{3}{2} \right) \right) + \left(-\frac{1}{2} - 0 \right) = \frac{3}{2} - \frac{1}{2} = 1 \end{aligned}$$

Example 19: The Beginner's Trap

$$\int_0^{3\pi/2} \sqrt{\sin^2 x} dx$$

A common mistake is evaluating this as simply $\int \sin x dx$. The square root of a square introduces an absolute value:

$$\int_0^{3\pi/2} |\sin x| dx$$

Split the bounds where $\sin x$ becomes negative (at $x = \pi$):

$$\int_0^\pi \sin x dx + \int_\pi^{3\pi/2} (-\sin x) dx = \left[-\cos x \right]_0^\pi + \left[\cos x \right]_\pi^{3\pi/2} = (1 - (-1)) + (0 - (-1)) = 2 + 1 = 3$$

Example 20

$$\int_0^3 |x^2 - 4| dx$$

Identify the root at $x = 2$ and split the domain:

$$\begin{aligned} \int_0^2 (4 - x^2) dx + \int_2^3 (x^2 - 4) dx &= \left[4x - \frac{x^3}{3} \right]_0^2 + \left[\frac{x^3}{3} - 4x \right]_2^3 \\ &= \left(8 - \frac{8}{3} \right) + \left((9 - 12) - \left(\frac{8}{3} - 8 \right) \right) = \frac{16}{3} + \left(-3 + \frac{16}{3} \right) = \frac{16}{3} + \frac{7}{3} = \frac{23}{3} \end{aligned}$$

Example 21

$$\int_0^4 |\sqrt{x} - 1| dx$$

Identify the root at $x = 1$ and split:

$$\begin{aligned} \int_0^1 (1 - \sqrt{x}) dx + \int_1^4 (\sqrt{x} - 1) dx &= \left[x - \frac{2}{3}x^{3/2} \right]_0^1 + \left[\frac{2}{3}x^{3/2} - x \right]_1^4 \\ &= \left(1 - \frac{2}{3} \right) + \left(\left(\frac{16}{3} - 4 \right) - \left(\frac{2}{3} - 1 \right) \right) = \frac{1}{3} + \left(\frac{4}{3} - \left(-\frac{1}{3} \right) \right) = \frac{1}{3} + \frac{5}{3} = 2 \end{aligned}$$

Chapter 2

Integration Bee Training — Part 2: U-Substitution

This section focuses on mastering the speed and mental detection of the core components of u -substitution: spotting the inner function u and its differential du . Mastery here allows you to bypass writing the formal substitution steps, granting immense speed advantages in competition.

2.1 Core Substitution Patterns

Example 1

$$\int x e^{-x^2} dx$$

Let $u = -x^2 \implies du = -2x dx \implies x dx = -\frac{1}{2} du$.

$$\int -\frac{1}{2} e^u du = -\frac{1}{2} e^{-x^2} + C$$

Example 2

$$\int \frac{x^3}{x^4 - 5} dx$$

Let $u = x^4 - 5 \implies du = 4x^3 dx$.

$$\frac{1}{4} \int \frac{1}{u} du = \frac{1}{4} \ln |u| + C = \frac{1}{4} \ln |x^4 - 5| + C$$

Example 3

$$\int \sqrt{2x - 1} dx$$

Let $u = 2x - 1 \implies du = 2 dx$.

$$\frac{1}{2} \int \sqrt{u} du = \frac{1}{2} \left(\frac{2}{3} u^{3/2} \right) + C = \frac{1}{3} (2x - 1)^{3/2} + C$$

Example 4

$$\int (1 - x)^9 dx$$

Let $u = 1 - x \implies du = -dx$.

$$-\int u^9 du = -\frac{u^{10}}{10} + C = -\frac{(1 - x)^{10}}{10} + C$$

Example 5

$$\int \frac{1}{5-3x} dx$$

Let $u = 5 - 3x \implies du = -3 dx$.

$$-\frac{1}{3} \int \frac{1}{u} du = -\frac{1}{3} \ln |5 - 3x| + C$$

Example 6

$$\int \cos^3 x \sin x dx$$

Let $u = \cos x \implies du = -\sin x dx$.

$$-\int u^3 du = -\frac{u^4}{4} + C = -\frac{\cos^4 x}{4} + C$$

Example 7

$$\int \frac{e^x}{1-e^x} dx$$

Let $u = 1 - e^x \implies du = -e^x dx$.

$$-\int \frac{1}{u} du = -\ln |1 - e^x| + C$$

Example 8

$$\int \frac{\sin(\sqrt{x})}{\sqrt{x}} dx$$

Let $u = \sqrt{x} \implies du = \frac{1}{2\sqrt{x}} dx$.

$$2 \int \sin(u) du = -2 \cos(u) + C = -2 \cos(\sqrt{x}) + C$$

Example 9

$$\int \frac{\ln^2 x}{x} dx$$

Let $u = \ln x \implies du = \frac{1}{x} dx$.

$$\int u^2 du = \frac{u^3}{3} + C = \frac{\ln^3 x}{3} + C$$

Example 10

$$\int \sin(\cos x) \sin x dx$$

Let $u = \cos x \implies du = -\sin x dx$.

$$-\int \sin u du = \cos(u) + C = \cos(\cos x) + C$$

2.2 Advanced Structural Manipulations

Example 11: The Phantom x

$$\int x\sqrt{x+2} \, dx$$

Let $u = x + 2 \implies du = dx$. We still have an x remaining. We use the back-substitution $x = u - 2$.

$$\begin{aligned} \int (u-2)\sqrt{u} \, du &= \int (u^{3/2} - 2u^{1/2}) \, du \\ &= \frac{2}{5}u^{5/2} - \frac{4}{3}u^{3/2} + C = \frac{2}{5}(x+2)^{5/2} - \frac{4}{3}(x+2)^{3/2} + C \end{aligned}$$

Example 12

$$\int (x^2 + 1)(x^3 + 3x)^4 \, dx$$

Let $u = x^3 + 3x \implies du = (3x^2 + 3) \, dx = 3(x^2 + 1) \, dx$.

$$\frac{1}{3} \int u^4 \, du = \frac{u^5}{15} + C = \frac{(x^3 + 3x)^5}{15} + C$$

Example 13

$$\int \frac{\sec^2 x}{\tan^2 x} \, dx$$

Let $u = \tan x \implies du = \sec^2 x \, dx$.

$$\int \frac{1}{u^2} \, du = -\frac{1}{u} + C = -\cot x + C$$

Example 14

$$\int \sin(2x - 1) \, dx$$

Let $u = 2x - 1 \implies du = 2 \, dx$.

$$\frac{1}{2} \int \sin(u) \, du = -\frac{1}{2} \cos(2x - 1) + C$$

Example 15

$$\int \frac{\csc^2 x}{\sqrt{\cot x}} \, dx$$

Let $u = \cot x \implies du = -\csc^2 x \, dx$.

$$-\int \frac{1}{\sqrt{u}} \, du = -2\sqrt{u} + C = -2\sqrt{\cot x} + C$$

Example 16

$$\int \frac{e^x}{e^x + 1} \, dx$$

Let $u = e^x + 1 \implies du = e^x \, dx$.

$$\int \frac{1}{u} \, du = \ln(e^x + 1) + C$$

Example 17: Implicit Double Angle

$$\int \frac{\sin(2x)}{1 + \cos^2 x} dx$$

Rather than expanding $\sin(2x) = 2 \sin x \cos x$, check the derivative of the denominator directly:
 let $u = 1 + \cos^2 x \implies du = -2 \cos x \sin x dx = -\sin(2x) dx$.

$$- \int \frac{1}{u} du = -\ln(1 + \cos^2 x) + C$$

Bonus Fact: The derivative of $\sin^2 x$ is exactly $\sin(2x)$.

Example 18: Cotangent Primitive

$$\int \cot x dx = \int \frac{\cos x}{\sin x} dx$$

Let $u = \sin x \implies du = \cos x dx$.

$$\int \frac{1}{u} du = \ln |\sin x| + C$$

Example 19

$$\int \frac{\cos(\ln x)}{x} dx$$

Let $u = \ln x \implies du = \frac{1}{x} dx$.

$$\int \cos(u) du = \sin(u) + C = \sin(\ln x) + C$$

Example 20: Missing Polynomial Degree

$$\int \frac{x}{x^4 + 1} dx$$

We cannot use $u = x^4$ since we only have x on top (we would need x^3). Instead, let $u = x^2 \implies du = 2x dx$. The denominator x^4 becomes u^2 .

$$\frac{1}{2} \int \frac{1}{u^2 + 1} du = \frac{1}{2} \arctan(u) + C = \frac{1}{2} \arctan(x^2) + C$$

Example 21: Algebraic Splitting

$$\int \frac{1+x}{1+x^2} dx$$

Split the fraction into two components:

$$\int \frac{1}{1+x^2} dx + \int \frac{x}{1+x^2} dx$$

The first is standard $\arctan x$. For the second, let $u = 1 + x^2 \implies du = 2x dx$.

$$\arctan x + \frac{1}{2} \ln(1 + x^2) + C$$

Example 22: Degree Mismatch

$$\int \frac{x^3}{\sqrt{x^2+1}} dx$$

Let $u = x^2 + 1 \implies du = 2x dx$. Split x^3 into $x^2 \cdot x$. Back-substitute $x^2 = u - 1$.

$$\begin{aligned} \frac{1}{2} \int \frac{u-1}{\sqrt{u}} du &= \frac{1}{2} \int (u^{1/2} - u^{-1/2}) du \\ &= \frac{1}{2} \left(\frac{2}{3} u^{3/2} - 2u^{1/2} \right) + C = \frac{1}{3} (x^2 + 1)^{3/2} - \sqrt{x^2 + 1} + C \end{aligned}$$

2.3 Definite Integrals and Boundary Dynamics**Example 23**

$$\int_0^1 \frac{1}{\sqrt[3]{1+7x}} dx$$

Let $u = 1+7x \implies du = 7 dx$. Transform the bounds: $x = 0 \implies u = 1$ and $x = 1 \implies u = 8$.

$$\frac{1}{7} \int_1^8 u^{-1/3} du = \frac{1}{7} \left[\frac{3}{2} u^{2/3} \right]_1^8 = \frac{3}{14} (8^{2/3} - 1^{2/3}) = \frac{3}{14} (4 - 1) = \frac{9}{14}$$

Example 24: Boundary Reversal

$$\int_1^2 \frac{e^{1/x}}{x^2} dx$$

Let $u = \frac{1}{x} \implies du = -\frac{1}{x^2} dx$. Transform bounds: $x = 1 \implies u = 1$ and $x = 2 \implies u = 1/2$. The bounds are reversed ($1 \rightarrow 1/2$), but the negative sign from the differential corrects this:

$$-\int_1^{1/2} e^u du = \int_{1/2}^1 e^u du = \left[e^u \right]_{1/2}^1 = e - \sqrt{e}$$

Example 25: Parity Recognition

$$\int_{-\pi/4}^{\pi/4} (x^3 + x^4 \tan x) dx$$

Observe the symmetry of the bounds. Check the function parity: x^3 is odd. The term $x^4 \tan x$ is (even) $\times$ (odd) = odd. Because the entire function is odd over a symmetric interval $[-a, a]$, the integral equates perfectly to:

$$0$$

Example 26: Back-Substitution with Definite Bounds

$$\int_1^2 x\sqrt{x-1} dx$$

Let $u = x - 1 \implies du = dx$ and $x = u + 1$. Bounds change to $0 \rightarrow 1$.

$$\int_0^1 (u+1)\sqrt{u} du = \int_0^1 (u^{3/2} + u^{1/2}) du = \left[\frac{2}{5} u^{5/2} + \frac{2}{3} u^{3/2} \right]_0^1 = \frac{2}{5} + \frac{2}{3} = \frac{16}{15}$$

Example 27: Absolute Symmetry Annihilation

$$\int_{-\pi}^{\pi} x^4 \sin x \, dx$$

Since x^4 is even and $\sin x$ is odd, the product is an odd function across symmetric boundaries. The result is immediately:

$$0$$

Example 28: Logarithmic Bounds

$$\int_1^e \frac{\sqrt{\ln x}}{x} \, dx$$

Let $u = \ln x \implies du = \frac{1}{x} dx$. Bounds change from $0 \rightarrow 1$.

$$\int_0^1 \sqrt{u} \, du = \left[\frac{2}{3} u^{3/2} \right]_0^1 = \frac{2}{3}$$

Example 29: Hidden Symmetry in Translation

$$\int_0^2 (x-1)e^{(x-1)^2} \, dx$$

Let $u = x - 1 \implies du = dx$. The bounds shift to $-1 \rightarrow 1$.

$$\int_{-1}^1 u e^{u^2} \, du$$

Observe that $u e^{u^2}$ is an odd function evaluated over $[-1, 1]$. The result is immediately:

$$0$$

Example 30: Natural Logarithmic Offset

$$\int_0^1 \frac{e^x}{1+e^x} \, dx$$

Let $u = e^x + 1 \implies du = e^x dx$. Bounds change to $2 \rightarrow e + 1$.

$$\int_2^{e+1} \frac{1}{u} \, du = \ln(e+1) - \ln(2)$$

2.4 The Area of a Partial Circle Concept**Example 31: Quarter Circle Identification**

$$\int_0^1 \sqrt{1-x^2} \, dx$$

Rather than using trigonometric substitution, recognize this as the equation of a semi-circle $y = \sqrt{R^2 - x^2}$ with radius $R = 1$. The bounds 0 to 1 indicate exactly one-quarter of the circle's total area.

$$A = \frac{\pi R^2}{4} = \frac{\pi(1)^2}{4} = \frac{\pi}{4}$$

Example 32: Large Scale Circle

$$\int_0^{11} \sqrt{121 - x^2} dx$$

Recognize $121 = 11^2$. This is a quarter circle of radius $R = 11$.

$$\frac{\pi(11)^2}{4} = \frac{121\pi}{4}$$

Example 33: Non-Standard Elliptical Circle

$$\int_{-2}^2 \sqrt{12 - 3x^2} dx$$

Factor out the 3 from the internal polynomial:

$$\int_{-2}^2 \sqrt{3(4 - x^2)} dx = \sqrt{3} \int_{-2}^2 \sqrt{4 - x^2} dx$$

This reveals a semi-circle of radius $R = 2$ evaluated fully from -2 to 2 , forming a half-circle.

$$\sqrt{3} \left(\frac{\pi(2)^2}{2} \right) = 2\pi\sqrt{3}$$

Example 34: Combined Odd Function and Geometry

$$\int_{-2}^2 (x + 1) \sqrt{4 - x^2} dx$$

Distribute the radical:

$$\int_{-2}^2 x \sqrt{4 - x^2} dx + \int_{-2}^2 \sqrt{4 - x^2} dx$$

The first integral is an odd function evaluated symmetrically, yielding 0. The second integral represents a perfect half-circle with radius $R = 2$.

$$0 + \frac{\pi(2)^2}{2} = 2\pi$$

Example 35: U-Substitution into a Circle

$$\int_0^1 x \sqrt{1 - x^4} dx$$

Let $u = x^2 \implies du = 2x dx$. The bounds $0 \rightarrow 1$ remain the same. The term x^4 becomes u^2 .

$$\frac{1}{2} \int_0^1 \sqrt{1 - u^2} du$$

The remaining integral is exactly a quarter-circle with radius $R = 1$.

$$\frac{1}{2} \left(\frac{\pi(1)^2}{4} \right) = \frac{\pi}{8}$$

2.5 Concluding Mixed Applications

Example 36: Symmetric Cancellation

$$\int_{-\pi/4}^{\pi/4} \frac{x^4 \tan x}{2 + \cos x} dx$$

Check for parity. x^4 is even. $\tan x$ is odd. The denominator $2 + \cos x$ is even. The complete product is an odd function over symmetric bounds. The answer is directly:

$$0$$

Example 37: Disguised Power Rule

$$\int (1 + x^{1/4})^2 \frac{1}{\sqrt{x}} dx$$

Expand the perfect square: $(1 + 2x^{1/4} + x^{1/2})$. Multiply by $x^{-1/2}$:

$$\int (x^{-1/2} + 2x^{-1/4} + 1) dx = 2x^{1/2} + \frac{8}{3}x^{3/4} + x + C$$

Example 38: Infinite Arctangent Limit

$$\int_0^\infty \frac{e^x}{1 + e^{2x}} dx$$

Let $u = e^x \implies du = e^x dx$. The bounds change from $0 \rightarrow \infty$ to $1 \rightarrow \infty$.

$$\int_1^\infty \frac{1}{1 + u^2} du = \left[\arctan(u) \right]_1^\infty = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

Example 39: Radical Polynomial Factoring

$$\int \frac{x + 2}{\sqrt{x^2 + 4x}} dx$$

Let $u = x^2 + 4x \implies du = (2x + 4) dx = 2(x + 2) dx$.

$$\frac{1}{2} \int \frac{1}{\sqrt{u}} du = \sqrt{u} + C = \sqrt{x^2 + 4x} + C$$

Example 40: Exponential Chain

$$\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$$

Let $u = \sqrt{x} \implies du = \frac{1}{2\sqrt{x}} dx$.

$$2 \int e^u du = 2e^u + C = 2e^{\sqrt{x}} + C$$

Example 41: Secant Logarithmic Integral

$$\int \frac{\sec x \tan x}{\sec x + 1} dx$$

Let $u = \sec x + 1 \implies du = \sec x \tan x dx$.

$$\int \frac{1}{u} du = \ln |\sec x + 1| + C$$

Chapter 3

Integration Bee Training — Part 3: Integration by Parts

This section focuses on mastering the foundational rules of integration by parts, including the DI (Tabular) method, reverse product rule, recursive integrals, and forced U-substitution. Mastery of these fundamentals is necessary before advancing to competitive speed integration techniques.

3.1 The DI (Tabular) Method

Example 1

$$\int x e^{2x} dx$$

Apply the DI method by differentiating x and integrating e^{2x} :

$$\int x e^{2x} dx = \frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} + C$$

Example 2

$$\int \sqrt{x} \ln(x) dx$$

Apply the DI method by differentiating $\ln(x)$ and integrating $\sqrt{x}$:

$$\int \sqrt{x} \ln(x) dx = \frac{2}{3} x^{3/2} \ln(x) - \frac{4}{9} x^{3/2} + C$$

Example 3

$$\int x^2 \cos(2x) dx$$

Apply the DI method by differentiating x^2 and integrating $\cos(2x)$:

$$\int x^2 \cos(2x) dx = \frac{1}{2} x^2 \sin(2x) + \frac{1}{2} x \cos(2x) - \frac{1}{4} \sin(2x) + C$$

Example 4

$$\int (x^2 + 2x) e^x dx$$

Apply the DI method, observing that the trailing terms perfectly cancel out:

$$\int (x^2 + 2x) e^x dx = x^2 e^x + C$$

3.2 Reverse Product Rule

Example 5

$$\int (5x^4 e^x + x^5 e^x) dx$$

Recognize the integrand as the exact derivative of a product:

$$\int (5x^4 e^x + x^5 e^x) dx = x^5 e^x + C$$

Example 6

$$\int (e^x \sin x + e^x \cos x) dx$$

Recognize the integrand as the exact derivative of $e^x \sin x$:

$$\int (e^x \sin x + e^x \cos x) dx = e^x \sin x + C$$

Example 7

$$\int (2x \sin x + x^2 \cos x) dx$$

Recognize the integrand as the exact derivative of $x^2 \sin x$:

$$\int (2x \sin x + x^2 \cos x) dx = x^2 \sin x + C$$

3.3 Definite Integrals and Algebraic Manipulations

Example 8

$$\int_0^1 \frac{x^2 + 1}{e^x} dx$$

Rewrite with a negative exponent and apply the DI method:

$$\int_0^1 (x^2 + 1)e^{-x} dx = 3 - \frac{6}{e}$$

3.4 Inverse Functions and Logarithms

Example 9

$$\int \arccos(x) dx$$

Apply integration by parts by differentiating $\arccos(x)$ and integrating 1:

$$\int \arccos(x) dx = x \arccos(x) - \sqrt{1 - x^2} + C$$

Example 10

$$\int \ln(\sqrt{x}) dx$$

Simplify using logarithm properties before integrating:

$$\int \ln(\sqrt{x}) dx = \frac{1}{2}x \ln(x) - \frac{1}{2}x + C$$

Example 11

$$\int x^4 \ln(x) dx$$

Apply the DI method by differentiating $\ln(x)$ and integrating x^4 :

$$\int x^4 \ln(x) dx = \frac{x^5}{5} \ln(x) - \frac{x^5}{25} + C$$

Example 12

$$\int \arctan(x) dx$$

Apply integration by parts by differentiating $\arctan(x)$ and integrating 1:

$$\int \arctan(x) dx = x \arctan(x) - \frac{1}{2} \ln(1 + x^2) + C$$

Example 13

$$\int x \csc^2(x) dx$$

Apply the DI method by differentiating x and integrating $\csc^2(x)$:

$$\int x \csc^2(x) dx = -x \cot(x) + \ln |\sin(x)| + C$$

3.5 Forced U-Substitution**Example 14**

$$\int (\ln x)^4 dx$$

Force the substitution $u = \ln x$, yielding $dx = e^u du$:

$$\int (\ln x)^4 dx = x ((\ln x)^4 - 4(\ln x)^3 + 12(\ln x)^2 - 24 \ln x + 24) + C$$

Example 15

$$\int_0^1 (\arcsin x)^2 dx$$

Force the substitution $u = \arcsin x$ to change the bounds and integrand:

$$\int_0^1 (\arcsin x)^2 dx = \frac{\pi^2}{4} - 2$$

Example 16

$$\int e^{\sqrt{x}} dx$$

Force the substitution $u = \sqrt{x}$, yielding $dx = 2u du$:

$$\int e^{\sqrt{x}} dx = 2\sqrt{x}e^{\sqrt{x}} - 2e^{\sqrt{x}} + C$$

Example 17

$$\int \cos(\ln x) dx$$

Force the substitution $u = \ln x$, yielding $dx = e^u du$:

$$\int \cos(\ln x) dx = \frac{x}{2} (\sin(\ln x) + \cos(\ln x)) + C$$

Example 18

$$\int \sin(\sqrt{x}) dx$$

Force the substitution $u = \sqrt{x}$, yielding $dx = 2u du$:

$$\int \sin(\sqrt{x}) dx = -2\sqrt{x} \cos(\sqrt{x}) + 2 \sin(\sqrt{x}) + C$$

3.6 Recursive Integration and Advanced Techniques**Example 19**

$$\int e^{2x} \sin(3x) dx$$

Apply the DI method twice, set equal to I , and solve algebraically:

$$\int e^{2x} \sin(3x) dx = \frac{e^{2x}}{13} (2 \sin(3x) - 3 \cos(3x)) + C$$

Example 20

$$\int x \sin x \cos x dx$$

Convert to $\frac{1}{2} \sin(2x)$ before applying the DI method:

$$\int x \sin x \cos x dx = \frac{1}{8} \sin(2x) - \frac{x}{4} \cos(2x) + C$$

Example 21

$$\int_1^{\sqrt{3}} \arctan\left(\frac{1}{x}\right) dx$$

Apply the DI method by differentiating $\arctan(1/x)$ and integrating 1:

$$\int_1^{\sqrt{3}} \arctan\left(\frac{1}{x}\right) dx = \frac{\pi\sqrt{3}}{6} + \frac{1}{2} \ln(2) - \frac{\pi}{4}$$

Example 22

$$\int_0^{\pi/3} \sin x \ln(\cos x) dx$$

Apply the DI method by differentiating $\ln(\cos x)$ and integrating $\sin x$:

$$\int_0^{\pi/3} \sin x \ln(\cos x) dx = \frac{1}{2} \ln(2) - \frac{1}{2}$$

Example 23: Zero Substitution

$$\int x \ln(x+1) dx$$

Use the zero substitution trick by adding and subtracting 1 and x to divide the polynomial smoothly:

$$\int x \ln(x+1) dx = \frac{x^2}{2} \ln(x+1) - \frac{1}{2} \left(\frac{x^2}{2} - x + \ln(x+1) \right) + C$$

Chapter 4

Integration Bee Training — Part 4: Trigonometric Integrals

This section focuses on manipulating trigonometric integrals using identities, U-substitutions, Wallis's product, product-to-sum formulas, and recognizing symmetry. Mastery of these techniques is essential for bypassing lengthy algebraic methods during speed integration.

4.1 Sines, Cosines, and U-Substitutions

Example 1

$$\int \sin^2(x) \cos^3(x) dx$$

Factor out one $\cos(x)$ and convert the remaining $\cos^2(x)$ to $1 - \sin^2(x)$, then use the substitution $u = \sin(x)$:

$$\int \sin^2(x)(1 - \sin^2(x)) \cos(x) dx = \int (u^2 - u^4) du = \frac{\sin^3(x)}{3} - \frac{\sin^5(x)}{5} + C$$

Example 2: Wallis's Product (Odd Power)

$$\int_0^{\pi/2} \sin^5(x) dx$$

Using Wallis's trick for odd powers, continuously multiply the sequence of fractions starting from the bottom up to the power n :

$$\int_0^{\pi/2} \sin^5(x) dx = \frac{2 \cdot 4}{3 \cdot 5} = \frac{8}{15}$$

Example 3

$$\int x \cos^5(x^2) dx$$

Let $u = x^2$, making $du = 2x dx$. The integral becomes $\frac{1}{2} \int \cos^5(u) du$. Factor out $\cos(u)$, convert $\cos^4(u)$ to $(1 - \sin^2(u))^2$, and substitute $v = \sin(u)$:

$$\frac{1}{2} \int (1 - v^2)^2 dv = \frac{1}{2} \left(v - \frac{2v^3}{3} + \frac{v^5}{5} \right) = \frac{\sin(x^2)}{2} - \frac{\sin^3(x^2)}{3} + \frac{\sin^5(x^2)}{10} + C$$

Example 4

$$\int \sin^2(x) dx$$

Apply the half-angle identity $\sin^2(x) = \frac{1 - \cos(2x)}{2}$:

$$\int \frac{1 - \cos(2x)}{2} dx = \frac{x}{2} - \frac{\sin(2x)}{4} + C$$

Example 5

$$\int \cos^4(x) dx$$

Apply the half-angle identity twice. First, convert $\cos^2(x)$ to $\frac{1 + \cos(2x)}{2}$, and square it. Apply the identity again to the resulting $\cos^2(2x)$:

$$\int \left(\frac{1}{4} + \frac{\cos(2x)}{2} + \frac{1 + \cos(4x)}{8} \right) dx = \frac{3x}{8} + \frac{\sin(2x)}{4} + \frac{\sin(4x)}{32} + C$$

Example 6

$$\int \sin^2(x) \cos^2(x) dx$$

Recognize $(\sin(x) \cos(x))^2 = \frac{\sin^2(2x)}{4}$. Apply the half-angle identity to $\sin^2(2x)$:

$$\int \frac{1 - \cos(4x)}{8} dx = \frac{x}{8} - \frac{\sin(4x)}{32} + C$$

Example 7

$$\int \sin^3(x) \cos^5(x) dx$$

Factor out $\cos(x)$, substitute $u = \sin(x)$, and convert the remaining $\cos^4(x)$ to $(1 - u^2)^2$:

$$\int u^3(1 - u^2)^2 du = \int (u^3 - 2u^5 + u^7) du = \frac{\sin^4(x)}{4} - \frac{\sin^6(x)}{3} + \frac{\sin^8(x)}{8} + C$$

Example 8

$$\int_0^{\pi/2} (2 - \sin x)^2 dx$$

Expand to $4 - 4\sin(x) + \sin^2(x)$. Integrate each part, using Wallis's trick for the $\sin^2(x)$ term:

$$[4x + 4\cos(x)]_0^{\pi/2} + \frac{\pi}{4} = (2\pi - 0) + (0 - 4) + \frac{\pi}{4} = \frac{9\pi}{4} - 4$$

4.2 Secants and Tangents**Example 9**

$$\int \sec^3(x) \tan(x) dx$$

Let $u = \sec(x)$, so $du = \sec(x) \tan(x) dx$. The integral reduces smoothly:

$$\int u^2 du = \frac{\sec^3(x)}{3} + C$$

Example 10

$$\int \sec^6(x) \tan^4(x) dx$$

Factor out $\sec^2(x) dx$, convert the remaining $\sec^4(x)$ to $(\tan^2(x)+1)^2$, and substitute $u = \tan(x)$:

$$\int u^4(u^2 + 1)^2 du = \int (u^8 + 2u^6 + u^4) du = \frac{\tan^9(x)}{9} + \frac{2 \tan^7(x)}{7} + \frac{\tan^5(x)}{5} + C$$

Example 11

$$\int \tan^3(x) \sec(x) dx$$

Factor out $\sec(x) \tan(x) dx$, convert the remaining $\tan^2(x)$ to $\sec^2(x) - 1$, and let $u = \sec(x)$:

$$\int (u^2 - 1) du = \frac{\sec^3(x)}{3} - \sec(x) + C$$

Example 12

$$\int \tan^2(x) \cos^3(x) dx$$

Convert everything to sines and cosines to find $\frac{\sin^2(x)}{\cos^2(x)} \cos^3(x) = \sin^2(x) \cos(x)$. Let $u = \sin(x)$:

$$\int u^2 du = \frac{\sin^3(x)}{3} + C$$

Example 13

$$\int (\tan^2(x) + \tan^4(x)) dx$$

Factor out $\tan^2(x)$ to get $\tan^2(x)(1 + \tan^2(x)) = \tan^2(x) \sec^2(x)$. Let $u = \tan(x)$:

$$\int u^2 du = \frac{\tan^3(x)}{3} + C$$

Example 14

$$\int \tan^4(x) dx$$

Convert a $\tan^2(x)$ to $\sec^2(x) - 1$. Expand and apply the identity again on the remaining squared tangent:

$$\int (\tan^2(x) \sec^2(x) - \sec^2(x) + 1) dx = \frac{\tan^3(x)}{3} - \tan(x) + x + C$$

Example 15

$$\int \tan^5(x) dx$$

Peel off a factor of $\tan(x) \sec^2(x)$ and repeatedly apply $\tan^2(x) = \sec^2(x) - 1$:

$$\int (\tan^3(x) \sec^2(x) - \tan^3(x)) dx = \frac{\tan^4(x)}{4} - \frac{\tan^2(x)}{2} - \ln |\cos(x)| + C$$

4.3 Product-to-Sum and Advanced Manipulations

Example 16

$$\int \sin(4x) \cos(5x) dx$$

Apply the product-to-sum identity $\sin(A) \cos(B) = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$:

$$\frac{1}{2} \int (\sin(9x) - \sin(x)) dx = -\frac{\cos(9x)}{18} + \frac{\cos(x)}{2} + C$$

Example 17

$$\int \cos(2x) \cos(3x) dx$$

Apply the product-to-sum identity $\cos(A) \cos(B) = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$:

$$\frac{1}{2} \int (\cos(5x) + \cos(x)) dx = \frac{\sin(5x)}{10} + \frac{\sin(x)}{2} + C$$

Example 18

$$\int \sin(6x) \sin(8x) dx$$

Apply the product-to-sum identity $\sin(A) \sin(B) = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$:

$$\frac{1}{2} \int (\cos(2x) - \cos(14x)) dx = \frac{\sin(2x)}{4} - \frac{\sin(14x)}{28} + C$$

Example 19

$$\int_0^{\pi/6} \sqrt{1 + \cos(2x)} dx$$

Use the identity $1 + \cos(2x) = 2 \cos^2(x)$. Because of the square root, use absolute value $|\cos(x)|$:

$$\int_0^{\pi/6} \sqrt{2} |\cos(x)| dx = \sqrt{2} [\sin(x)]_0^{\pi/6} = \frac{\sqrt{2}}{2}$$

Example 20

$$\int \frac{1 - \tan^2(x)}{1 + \tan^2(x)} dx$$

Recognize the denominator is $\sec^2(x)$. Convert completely to sines and cosines: $\cos^2(x) - \sin^2(x) = \cos(2x)$:

$$\int \cos(2x) dx = \frac{\sin(2x)}{2} + C$$

Example 21: Trig Conjugation

$$\int \frac{1}{1 - \cos(x)} dx$$

Multiply the numerator and denominator by the conjugate $1 + \cos(x)$. The denominator becomes $1 - \cos^2(x) = \sin^2(x)$:

$$\int \frac{1 + \cos(x)}{\sin^2(x)} dx = \int (\csc^2(x) + \cot(x) \csc(x)) dx = -\cot(x) - \csc(x) + C$$

4.4 Symmetry Traps and Memorization

Example 22

$$\int_{-\pi/2}^{\pi/2} \sin(2x) \cos(x) dx$$

Before attempting any product-to-sum identities, check the bounds. The integrand is an odd function over symmetric bounds, so the integral evaluates to zero:

$$\int_{-\pi/2}^{\pi/2} \sin(2x) \cos(x) dx = 0$$

Example 23

$$\int \sec^3(x) dx$$

This standard recursive integration by parts problem should be solved purely by memory in a speed competition:

$$\int \sec^3(x) dx = \frac{1}{2}(\sec(x) \tan(x) + \ln |\sec(x) + \tan(x)|) + C$$

Example 24: Symmetry Trap (Odd Power)

$$\int_{-\pi}^{\pi} \sin^3(x) dx$$

Do not default to expanding the integral. Always notice symmetric bounds combined with an odd function:

$$\int_{-\pi}^{\pi} \sin^3(x) dx = 0$$

Chapter 5

Advanced Integration Training — Chapter: Trig-Substitution

This summary covers the examples from the Trigonometric Substitution chapter, demonstrating methods for simplifying radical integrands using standard identities or alternative algebraic substitutions.

5.1 Examples and Solutions

Example 1: Basic Sine Substitution

Evaluate the indefinite integral:

$$\int \frac{x^2}{\sqrt{1-x^2}} dx$$

Let $x = \sin \theta$, which implies $dx = \cos \theta d\theta$ and $\theta = \sin^{-1}(x)$.

$$\begin{aligned} \int \frac{x^2}{\sqrt{1-x^2}} dx &= \int \frac{\sin^2 \theta}{\cos \theta} \cdot \cos \theta d\theta \\ &= \int \sin^2 \theta d\theta \\ &= \int \left(\frac{1}{2} - \frac{1}{2} \cos(2\theta) \right) d\theta \\ &= \frac{\theta}{2} - \frac{1}{4} \sin(2\theta) + C \\ &= \frac{1}{2} \theta - \frac{1}{2} \sin \theta \cos \theta + C \end{aligned}$$

Using the reference triangle where $\sin \theta = x$ and $\cos \theta = \sqrt{1-x^2}$, we get:

$$\frac{1}{2} \sin^{-1}(x) - \frac{1}{2} x \sqrt{1-x^2} + C$$

Example 2: Secant Substitution with Powers

Evaluate the indefinite integral:

$$\int \frac{\sqrt{x^2-1}}{x^4} dx$$

Let $x = \sec \theta$, which implies $dx = \sec \theta \tan \theta d\theta$.

$$\begin{aligned}\int \frac{\tan \theta}{\sec^4 \theta} \cdot \sec \theta \tan \theta d\theta &= \int \frac{\tan^2 \theta}{\sec^3 \theta} d\theta \\ &= \int \sin^2 \theta \cos \theta d\theta\end{aligned}$$

Let $u = \sin \theta$, so $du = \cos \theta d\theta$:

$$\int u^2 du = \frac{1}{3}u^3 + C = \frac{1}{3}\sin^3 \theta + C$$

From the reference triangle where $\sec \theta = x$, we have $\sin \theta = \frac{\sqrt{x^2-1}}{x}$. Substituting back gives:

$$\frac{(x^2 - 1)^{3/2}}{3x^3} + C$$

Example 3: Direct Algebraic u -Substitution Alternative

$$\int_0^6 \frac{x}{\sqrt{36 - x^2}} dx$$

Instead of trigonometric substitution, an algebraic substitution works more directly. Let $u = 36 - x^2$, so $du = -2x dx \implies x dx = -\frac{1}{2}du$. Changing the limits: when $x = 0$, $u = 36$; when $x = 6$, $u = 0$.

$$\int_{36}^0 -\frac{1}{2} \frac{du}{\sqrt{u}} = \int_0^{36} \frac{du}{2\sqrt{u}}$$

Evaluating the foundational radical integral:

$$\left[\sqrt{u} \right]_0^{36} = \sqrt{36} - 0 = 6$$

Example 4: Secant Substitution with Simplification

Evaluate the indefinite integral:

$$\int \frac{dx}{x^2 \sqrt{x^2 - 1}}$$

Let $x = \sec \theta$, so $dx = \sec \theta \tan \theta d\theta$.

$$\int \frac{\sec \theta \tan \theta d\theta}{\sec^2 \theta \tan \theta} = \int \frac{1}{\sec \theta} d\theta = \int \cos \theta d\theta = \sin \theta + C$$

Using the reference triangle where $\sec \theta = x$, we have $\sin \theta = \frac{\sqrt{x^2-1}}{x}$:

$$\frac{\sqrt{x^2 - 1}}{x} + C$$

Example 5: Higher-Power Radical Identity

Evaluate the indefinite integral:

$$\int \frac{dx}{(x^2 - 1)^{3/2}}$$

Let $x = \sec \theta$, so $dx = \sec \theta \tan \theta d\theta$ and $\sqrt{x^2 - 1} = \tan \theta$.

$$\int \frac{\sec \theta \tan \theta d\theta}{\tan^3 \theta} = \int \frac{\sec \theta}{\tan^2 \theta} d\theta = \int \frac{\cos \theta}{\sin^2 \theta} d\theta$$

Let $u = \sin \theta$, so $du = \cos \theta d\theta$:

$$\int u^{-2} du = -\frac{1}{u} + C = -\csc \theta + C$$

Using the reference triangle where $\csc \theta = \frac{x}{\sqrt{x^2-1}}$:

$$-\frac{x}{\sqrt{x^2-1}} + C$$

Example 6: Scaling and Geometric Interpretation

$$\int_0^{2/3} \sqrt{4-9x^2} dx$$

Factor out the constant from the radical to match a standard circular curve form:

$$\int_0^{2/3} \sqrt{9\left(\frac{4}{9}-x^2\right)} dx = 3 \int_0^{2/3} \sqrt{\frac{4}{9}-x^2} dx$$

The integral $\int_0^{2/3} \sqrt{\frac{4}{9}-x^2} dx$ represents the area under a quarter circle centered at the origin with radius $r = \frac{2}{3}$ from $x = 0$ to $x = \frac{2}{3}$.

$$3 \cdot \left(\frac{1}{4}\pi r^2\right) = 3 \cdot \left(\frac{\pi\left(\frac{4}{9}\right)}{4}\right) = \frac{\pi}{3}$$

Example 7: Basic Tangent Substitution

Evaluate the indefinite integral:

$$\int \frac{dx}{\sqrt{4+x^2}}$$

Let $x = 2 \tan \theta$, which means $\tan \theta = \frac{x}{2}$ and $dx = 2 \sec^2 \theta d\theta$. From the reference triangle, $\sec \theta = \frac{\sqrt{4+x^2}}{2}$.

$$\begin{aligned} \int \frac{2 \sec^2 \theta d\theta}{2 \sec \theta} &= \int \sec \theta d\theta \\ &= \ln |\sec \theta + \tan \theta| + C \\ &= \ln \left| \frac{\sqrt{4+x^2}}{2} + \frac{x}{2} \right| + C \\ &= \ln |\sqrt{4+x^2} + x| + C_1 \end{aligned}$$

Example 8: Definite Secant Substitution with Powers

$$\int_1^{\sqrt{2}} \frac{\sqrt{x^2-1}}{x^3} dx$$

Let $x = \sec \theta$, so $dx = \sec \theta \tan \theta d\theta$. Change the limits: when $x = 1$, $\theta = 0$; when $x = \sqrt{2}$, $\theta = \frac{\pi}{4}$.

$$\begin{aligned} \int_0^{\pi/4} \frac{\tan \theta \cdot \sec \theta \tan \theta d\theta}{\sec^3 \theta} &= \int_0^{\pi/4} \frac{\tan^2 \theta}{\sec^2 \theta} d\theta \\ &= \int_0^{\pi/4} \sin^2 \theta d\theta \\ &= \int_0^{\pi/4} \left(\frac{1}{2} - \frac{1}{2} \cos(2\theta) \right) d\theta \\ &= \left[\frac{\theta}{2} - \frac{1}{4} \sin(2\theta) \right]_0^{\pi/4} \\ &= \frac{\pi}{8} - \frac{1}{4} \end{aligned}$$

Example 9: Definite Integration with Power of Tangent Form

$$\int_0^1 \frac{dx}{(1+x^2)^2}$$

Let $x = \tan \theta$, so $dx = \sec^2 \theta d\theta$. Change the limits: when $x = 0$, $\theta = 0$; when $x = 1$, $\theta = \frac{\pi}{4}$.

$$\begin{aligned} \int_0^{\pi/4} \frac{\sec^2 \theta d\theta}{\sec^4 \theta} &= \int_0^{\pi/4} \cos^2 \theta d\theta \\ &= \int_0^{\pi/4} \left(\frac{1}{2} + \frac{1}{2} \cos(2\theta) \right) d\theta \\ &= \left[\frac{\theta}{2} + \frac{1}{4} \sin(2\theta) \right]_0^{\pi/4} \\ &= \frac{\pi}{8} + \frac{1}{4} \end{aligned}$$

Example 10: Rational Tangent Substitution to u -Substitution

Evaluate the indefinite integral:

$$\int \frac{\sqrt{x^2+1}}{x} dx$$

Let $x = \tan \theta$, so $dx = \sec^2 \theta d\theta$.

$$\int \frac{\sec^3 \theta d\theta}{\tan \theta} = \int \frac{\sec^2 \theta}{\tan \theta} \sec \theta d\theta = \int \frac{\sec \theta}{\sin \theta \cos \theta} \cos \theta d\theta = \int \frac{\sec^2 \theta}{\tan \theta} d\theta$$

Alternatively, converting to sines and cosines:

$$\int \frac{1}{\cos^3 \theta} \cdot \frac{\cos \theta}{\sin \theta} d\theta = \int \frac{d\theta}{\sin \theta \cos^2 \theta} = \int \frac{\sin \theta}{\sin^2 \theta \cos^2 \theta} d\theta$$

Let $u = \cos \theta$, so $du = -\sin \theta d\theta$:

$$\int \frac{-du}{(1-u^2)u^2} = \int \left(-\frac{1}{u^2} - \frac{1}{1-u^2} \right) du = \frac{1}{u} - \frac{1}{2} \ln \left| \frac{1+u}{1-u} \right| + C$$

Substituting back $u = \cos \theta = \frac{1}{\sqrt{x^2+1}}$:

$$\sqrt{x^2+1} - \ln \left| \frac{\sqrt{x^2+1}+1}{x} \right| + C$$

Example 11: Completing the Square followed by Tangent Substitution

Evaluate the indefinite integral:

$$\int \frac{dx}{\sqrt{x^2 + 2x + 5}}$$

First, complete the square for the quadratic expression under the radical:

$$x^2 + 2x + 5 = (x + 1)^2 + 4$$

Let $x + 1 = 2 \tan \theta$, so $dx = 2 \sec^2 \theta d\theta$, and $\sec \theta = \frac{\sqrt{x^2 + 2x + 5}}{2}$.

$$\begin{aligned} \int \frac{2 \sec^2 \theta d\theta}{2 \sec \theta} &= \int \sec \theta d\theta \\ &= \ln |\sec \theta + \tan \theta| + C \\ &= \ln \left| \frac{\sqrt{x^2 + 2x + 5}}{2} + \frac{x + 1}{2} \right| + C \\ &= \ln \left| \sqrt{x^2 + 2x + 5} + x + 1 \right| + C_1 \end{aligned}$$

Example 12: Definite Integration by Completing the Square

$$\int_0^1 \sqrt{x - x^2} dx$$

First, complete the square for the quadratic expression under the radical:

$$x - x^2 = -(x^2 - x) = -\left(\left(x - \frac{1}{2}\right)^2 - \frac{1}{4}\right) = \frac{1}{4} - \left(x - \frac{1}{2}\right)^2$$

Rewrite the integral:

$$\int_0^1 \sqrt{\frac{1}{4} - \left(x - \frac{1}{2}\right)^2} dx$$

Let $x - \frac{1}{2} = \frac{1}{2} \sin \theta$, which gives $dx = \frac{1}{2} \cos \theta d\theta$.

Determine the new integration limits:

- When $x = 0$, $\frac{1}{2} \sin \theta = -\frac{1}{2} \implies \theta = -\frac{\pi}{2}$
- When $x = 1$, $\frac{1}{2} \sin \theta = \frac{1}{2} \implies \theta = \frac{\pi}{2}$

Substitute these into the integral:

$$\int_{-\pi/2}^{\pi/2} \sqrt{\frac{1}{4} - \frac{1}{4} \sin^2 \theta} \cdot \left(\frac{1}{2} \cos \theta\right) d\theta = \frac{1}{4} \int_{-\pi/2}^{\pi/2} \cos^2 \theta d\theta$$

By symmetry of the integrand:

$$\frac{2}{4} \int_0^{\pi/2} \cos^2 \theta d\theta = \frac{1}{2} \int_0^{\pi/2} \cos^2 \theta d\theta$$

Using the reduction formula $\int_0^{\pi/2} \cos^2 \theta d\theta = \frac{\pi}{4}$:

$$\frac{1}{2} \cdot \frac{\pi}{4} = \frac{\pi}{8}$$

Example 13: Completing the Square to Standard Inverse Sine Form

Evaluate the indefinite integral:

$$\int \frac{dx}{\sqrt{2x - x^2}}$$

Complete the square inside the radical:

$$2x - x^2 = 1 - (x - 1)^2$$

Substitute this back into the integral:

$$\int \frac{dx}{\sqrt{1 - (x - 1)^2}}$$

Let $u = x - 1$, so $du = dx$:

$$\int \frac{du}{\sqrt{1 - u^2}} = \sin^{-1}(u) + C = \sin^{-1}(x - 1) + C$$

Example 14: Completing the Square followed by Tangent Substitution

Evaluate the indefinite integral:

$$\int \frac{x^2}{(x^2 - 2x + 2)^2} dx$$

Complete the square for the denominator base: $x^2 - 2x + 2 = (x - 1)^2 + 1$.

$$\int \frac{x^2}{[(x - 1)^2 + 1]^2} dx$$

Let $x - 1 = \tan \theta$, which means $x = \tan \theta + 1$ and $dx = \sec^2 \theta d\theta$.

$$\int \frac{(\tan \theta + 1)^2 \sec^2 \theta}{\sec^4 \theta} d\theta = \int \frac{\tan^2 \theta + 2 \tan \theta + 1}{\sec^2 \theta} d\theta$$

Since $\tan^2 \theta + 1 = \sec^2 \theta$:

$$\int \frac{\sec^2 \theta + 2 \tan \theta}{\sec^2 \theta} d\theta = \int (1 + 2 \sin \theta \cos \theta) d\theta = \int (1 + \sin(2\theta)) d\theta$$

Integrating yields:

$$\theta - \frac{1}{2} \cos(2\theta) + C$$

Substituting back $\theta = \tan^{-1}(x - 1)$:

$$\tan^{-1}(x - 1) - \frac{1}{2} \cos(2 \tan^{-1}(x - 1)) + C$$

Example 15: Algebraic Secant Substitution with High Powers

Evaluate the indefinite integral:

$$\int \frac{dx}{x^4 \sqrt{x^2 - 1}}$$

Let $x = \sec \theta$, so $dx = \sec \theta \tan \theta d\theta$.

$$\int \frac{\sec \theta \tan \theta}{\sec^4 \theta \sqrt{\sec^2 \theta - 1}} d\theta = \int \frac{\sec \theta \tan \theta}{\sec^4 \theta \tan \theta} d\theta = \int \cos^3 \theta d\theta$$

Separate the terms: $\cos^3 \theta = \cos \theta (1 - \sin^2 \theta)$. Let $u = \sin \theta$, $du = \cos \theta d\theta$:

$$\int (1 - u^2) du = u - \frac{1}{3}u^3 + C = \sin \theta - \frac{1}{3}\sin^3 \theta + C$$

Substituting back $\theta = \sec^{-1}(x)$:

$$\sin(\sec^{-1}(x)) - \frac{1}{3}\sin^3(\sec^{-1}(x)) + C$$

Example 16: Definite Radical Integration via Scaling and Secant Substitution

$$\int_{3/2}^{3\sqrt{2}/2} \frac{\sqrt{4x^2 - 9}}{x} dx$$

Factor out the constant from the radical:

$$2 \int_{3/2}^{3\sqrt{2}/2} \frac{\sqrt{x^2 - \frac{9}{4}}}{x} dx$$

Let $x = \frac{3}{2} \sec \theta$, so $dx = \frac{3}{2} \sec \theta \tan \theta d\theta$.

- When $x = \frac{3}{2} \implies \sec \theta = 1 \implies \theta = 0$
- When $x = \frac{3\sqrt{2}}{2} \implies \sec \theta = \sqrt{2} \implies \theta = \frac{\pi}{4}$

Substituting into the expression:

$$2 \int_0^{\pi/4} \frac{\frac{3}{2} \tan \theta}{\frac{3}{2} \sec \theta} \cdot \left(\frac{3}{2} \sec \theta \tan \theta \right) d\theta = 3 \int_0^{\pi/4} \tan^2 \theta d\theta$$

Using $\tan^2 \theta = \sec^2 \theta - 1$:

$$3 \left[\tan \theta - \theta \right]_0^{\pi/4} = 3 \left(1 - \frac{\pi}{4} \right) = 3 - \frac{3\pi}{4}$$

Example 17: Rational Rational Integration via u -Substitution

$$\int_0^{1/\sqrt{2}} \frac{x^3}{(1 - x^2)^2} dx$$

Let $u = 1 - x^2$, which implies $x^2 = 1 - u$ and $du = -2x dx \implies x dx = -\frac{1}{2} du$.

- When $x = 0 \implies u = 1$
- When $x = \frac{1}{\sqrt{2}} \implies u = 1 - \frac{1}{2} = \frac{1}{2}$

Rewrite the integral:

$$\int_1^{1/2} \frac{1 - u}{u^2} \left(-\frac{1}{2} \right) du = \frac{1}{2} \int_{1/2}^1 \left(\frac{1}{u^2} - \frac{1}{u} \right) du = \frac{1}{2} \left[-\frac{1}{u} - \ln |u| \right]_{1/2}^1$$

Evaluating at the boundaries:

$$\frac{1}{2} \left[(-1 - 0) - \left(-2 - \ln \left(\frac{1}{2} \right) \right) \right] = \frac{1}{2} \left(1 + \ln \left(\frac{1}{2} \right) \right) = \frac{1}{2} + \frac{1}{2} \ln \left(\frac{1}{2} \right)$$

Example 18: Trigonometric Substitution for Composite Radical Fractions

$$\int_0^1 \frac{\sqrt{1-x}}{1-\sqrt{x}} dx$$

Let $x = \sin^2 \theta$, so $dx = 2 \sin \theta \cos \theta d\theta$. The limits transform: $x = 0 \rightarrow \theta = 0$ and $x = 1 \rightarrow \theta = \frac{\pi}{2}$.

$$\int_0^{\pi/2} \frac{\cos \theta}{1 - \sin \theta} \cdot (2 \sin \theta \cos \theta) d\theta = \int_0^{\pi/2} \frac{2 \sin \theta (1 - \sin^2 \theta)}{1 - \sin \theta} d\theta$$

Factoring the difference of squares:

$$\int_0^{\pi/2} 2 \sin \theta (1 + \sin \theta) d\theta = \int_0^{\pi/2} (2 \sin \theta + 2 \sin^2 \theta) d\theta$$

Evaluate each integrated part over the intervals:

$$\left[-2 \cos \theta \right]_0^{\pi/2} + \int_0^{\pi/2} (1 - \cos(2\theta)) d\theta = 2 + \left[\theta - \frac{1}{2} \sin(2\theta) \right]_0^{\pi/2} = 2 + \frac{\pi}{2}$$

Example 19: Inverse Cosine Integration via Double Angle Substitution

$$\int_0^1 \cos^{-1}(2x^2 - 1) dx$$

Let $x = \cos \theta$, so $dx = -\sin \theta d\theta$. Limits transform: $x = 0 \rightarrow \theta = \frac{\pi}{2}$ and $x = 1 \rightarrow \theta = 0$.

$$\int_{\pi/2}^0 \cos^{-1}(2 \cos^2 \theta - 1) \cdot (-\sin \theta) d\theta = \int_0^{\pi/2} \cos^{-1}(\cos(2\theta)) \sin \theta d\theta$$

Since $2\theta \in [0, \pi]$ over this domain, $\cos^{-1}(\cos(2\theta)) = 2\theta$:

$$\int_0^{\pi/2} 2\theta \sin \theta d\theta$$

Integrate by parts with $u = 2\theta$ and $dv = \sin \theta d\theta$:

$$\left[-2\theta \cos \theta \right]_0^{\pi/2} - \int_0^{\pi/2} (-2 \cos \theta) d\theta = \left[-2\theta \cos \theta + 2 \sin \theta \right]_0^{\pi/2} = 2$$

Chapter 6

Integration Bee Training — Part 6: Rational Polynomials

This section focuses on rapidly integrating rational functions using zero substitution, the Heaviside cover-up method, completing the square to force inverse tangents, and algebraic manipulations to bypass tedious standard partial fractions.

6.1 Zero Substitution and Algebraic Manipulation

Example 1

$$\int \frac{x^3}{x-1} dx$$

Use a zero substitution by borrowing a 1: subtract 1 and add 1. This allows factoring via the difference of cubes:

$$\int \frac{x^3 - 1 + 1}{x-1} dx = \int \left(x^2 + x + 1 + \frac{1}{x-1} \right) dx = \frac{x^3}{3} + \frac{x^2}{2} + x + \ln|x-1| + C$$

Example 2

$$\int \frac{3x-2}{x+1} dx$$

Force the numerator to match the denominator by adding and subtracting 5:

$$\int \frac{3x+3-5}{x+1} dx = \int \left(3 - \frac{5}{x+1} \right) dx = 3x - 5 \ln|x+1| + C$$

Example 3

$$\int \frac{2x-2}{x^3-x} dx$$

Factor the denominator and cancel out the common $(x-1)$ term:

$$\int \frac{2(x-1)}{x(x-1)(x+1)} dx = \int \frac{2}{x(x+1)} dx$$

By observation or basic partial fractions:

$$2 \int \left(\frac{1}{x} - \frac{1}{x+1} \right) dx = 2 \ln|x| - 2 \ln|x+1| + C$$

6.2 The Heaviside Cover-Up Method

Example 4

$$\int \frac{x-4}{(x-2)(x-3)} dx$$

Set up partial fractions $\frac{A}{x-2} + \frac{B}{x-3}$. Cover up the root and plug it into the remaining expression: For A (plug $x = 2$): $\frac{2-4}{2-3} = 2$. For B (plug $x = 3$): $\frac{3-4}{3-2} = -1$.

$$\int \left(\frac{2}{x-2} - \frac{1}{x-3} \right) dx = 2 \ln|x-2| - \ln|x-3| + C$$

Example 5

$$\int \frac{1}{(x+1)(x+2)(x+4)} dx$$

Apply the Heaviside method to each root (-1, -2, -4):

$$\int \left(\frac{1/3}{x+1} - \frac{1/2}{x+2} + \frac{1/6}{x+4} \right) dx = \frac{1}{3} \ln|x+1| - \frac{1}{2} \ln|x+2| + \frac{1}{6} \ln|x+4| + C$$

Example 6

$$\int \frac{1}{(x+1)^2(x+2)} dx$$

Set up $\frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x+2}$. Use Heaviside for the highest powers: $B = 1$ (plugging $x = -1$), $C = 1$ (plugging $x = -2$). Solve for A by picking a convenient x , like $x = 0$, yielding $A = -1$:

$$\int \left(\frac{-1}{x+1} + \frac{1}{(x+1)^2} + \frac{1}{x+2} \right) dx = -\ln|x+1| - \frac{1}{x+1} + \ln|x+2| + C$$

6.3 Completing the Square and Forcing Derivatives

Example 7

$$\int \frac{1}{x^2 + x + 1} dx$$

Complete the square in the denominator: $x^2 + x + 1 = (x + 1/2)^2 + 3/4$:

$$\int \frac{1}{(x + 1/2)^2 + 3/4} dx = \frac{2}{\sqrt{3}} \arctan \left(\frac{2x + 1}{\sqrt{3}} \right) + C$$

Example 8

$$\int \frac{x+4}{x^2 + 2x + 5} dx$$

The derivative of the denominator is $2x + 2$. Rewrite the numerator to isolate this:

$$\frac{1}{2} \int \frac{2x+2}{x^2 + 2x + 5} dx + \int \frac{3}{(x+1)^2 + 4} dx = \frac{1}{2} \ln(x^2 + 2x + 5) + \frac{3}{2} \arctan \left(\frac{x+1}{2} \right) + C$$

Example 9

$$\int \frac{x}{x^2 + 4x + 5} dx$$

The derivative of the denominator is $2x + 4$. Rewrite the numerator to match:

$$\frac{1}{2} \int \frac{2x + 4 - 4}{x^2 + 4x + 5} dx = \frac{1}{2} \ln(x^2 + 4x + 5) - 2 \arctan(x + 2) + C$$

6.4 Advanced Manipulations and Substitutions**Example 10: “Alpha” Substitution for Even Powers**

$$\int \frac{1}{x^4 - 16} dx$$

Instead of a massive partial fraction expansion, let $\alpha = x^2$ strictly for algebraic factoring: $\frac{1}{\alpha^2 - 16} = \frac{1}{(\alpha - 4)(\alpha + 4)}$. Apply Heaviside to get $\frac{1/8}{\alpha - 4} - \frac{1/8}{\alpha + 4}$:

$$\frac{1}{8} \int \left(\frac{1}{x^2 - 4} - \frac{1}{x^2 + 4} \right) dx = -\frac{1}{16} \operatorname{artanh}\left(\frac{x}{2}\right) - \frac{1}{16} \arctan\left(\frac{x}{2}\right) + C$$

Example 11: “Alpha” Substitution (Sum of Squares)

$$\int \frac{1}{1 - x^4} dx$$

Let $\alpha = x^2$, yielding $\frac{1}{(1 - \alpha)(1 + \alpha)} = \frac{1/2}{1 - \alpha} + \frac{1/2}{1 + \alpha}$. Substitute x^2 back in:

$$\frac{1}{2} \int \left(\frac{1}{1 - x^2} + \frac{1}{1 + x^2} \right) dx = \frac{1}{2} \operatorname{artanh}(x) + \frac{1}{2} \arctan(x) + C$$

Example 12

$$\int \frac{2x + 5}{(x^2 + 5x + 4)(x^2 + 5x + 6)} dx$$

Notice the derivative of $x^2 + 5x$ is $2x + 5$. Let $u = x^2 + 5x + 4$, so $du = (2x + 5)dx$. The integral becomes:

$$\int \frac{1}{u(u + 2)} du = \frac{1}{2} \int \left(\frac{1}{u} - \frac{1}{u + 2} \right) du = \frac{1}{2} \ln|x^2 + 5x + 4| - \frac{1}{2} \ln|x^2 + 5x + 6| + C$$

Example 13

$$\int \frac{e^x}{e^{2x} + 3e^x + 2} dx$$

Substitute $u = e^x$, yielding $\frac{1}{u^2 + 3u + 2} = \frac{1}{(u + 1)(u + 2)}$. Apply Heaviside:

$$\int \left(\frac{1}{u + 1} - \frac{1}{u + 2} \right) du = \ln(e^x + 1) - \ln(e^x + 2) + C$$

Example 14

$$\int \frac{\sin x}{\cos^2 x - 3 \cos x} dx$$

Substitute $u = \cos x$, $du = -\sin x dx$:

$$-\int \frac{1}{u(u - 3)} du = \int \left(\frac{1/3}{u} - \frac{1/3}{u - 3} \right) du = \frac{1}{3} \ln|\cos x| - \frac{1}{3} \ln|\cos x - 3| + C$$

6.5 Creative Zero Substitutions

Example 15

$$\int \frac{1}{x(x-2)^2} dx$$

Set up partial fractions $\frac{A}{x} + \frac{B}{x-2} + \frac{C}{(x-2)^2}$. Cover-up gives $A = 1/4$ (at $x = 0$) and $C = 1/2$ (at $x = 2$); matching the x^2 coefficient gives $B = -1/4$:

$$\int \left(\frac{1/4}{x} - \frac{1/4}{x-2} + \frac{1/2}{(x-2)^2} \right) dx = \frac{1}{4} \ln|x| - \frac{1}{4} \ln|x-2| - \frac{1}{2(x-2)} + C$$

Example 16

$$\int \frac{1}{x(x^2+1)} dx$$

Add and subtract x^2 in the numerator to manually split the fraction:

$$\int \frac{1+x^2-x^2}{x(x^2+1)} dx = \int \left(\frac{1}{x} - \frac{x}{x^2+1} \right) dx = \ln|x| - \frac{1}{2} \ln(x^2+1) + C$$

Chapter 7

Integration Bee Training — Part 7.1: Strategic Wildcard

This section focuses on compiling various strategies, throwing a "wildcard" of integrals that test your ability to quickly identify the best method—whether it be zero substitution, integration by parts, symmetry traps, or forced U-substitutions.

7.1 Clever Substitutions and Evaluating Constants

Example 1: Intimidating Constants

$$\int (x+1)^{\sqrt{2}} dx$$

Do not let the radical exponent intimidate you; it is strictly a constant. Let $u = x + 1$:

$$\int u^{\sqrt{2}} du = \frac{(x+1)^{\sqrt{2}+1}}{\sqrt{2}+1} + C$$

Example 2

$$\int e^{e^x+x} dx$$

Rewrite the exponent rules to factor the terms $\int e^{e^x} e^x dx$. Let $u = e^x$, $du = e^x dx$:

$$\int e^u du = e^{e^x} + C$$

Example 3

$$\int e^2 dx$$

Recognize immediately that e^2 contains no variable and is simply a constant:

$$\int e^2 dx = xe^2 + C$$

7.2 Algebraic Manipulations and Zero Substitutions

Example 4

$$\int_0^1 \frac{x}{(2x+1)^2} dx$$

Let $u = 2x + 1$, $dx = \frac{1}{2}du$, and $x = \frac{u-1}{2}$. Adjust the bounds from $[0, 1]$ to $[1, 3]$:

$$\frac{1}{4} \int_1^3 \frac{u-1}{u^2} du = \frac{1}{4} \int_1^3 \left(\frac{1}{u} - \frac{1}{u^2} \right) du = \frac{1}{4} \left[\ln|u| + \frac{1}{u} \right]_1^3 = \frac{1}{4} \left(\ln(3) - \frac{2}{3} \right)$$

Example 5

$$\int \frac{1+12x}{1+3x} dx$$

Use zero substitution to match the numerator with the denominator. Multiply the denominator by 4 to get $12x + 4$, so add and subtract 3:

$$\int \frac{4(1+3x) - 3}{1+3x} dx = \int \left(4 - \frac{3}{1+3x} \right) dx = 4x - \ln|1+3x| + C$$

Example 6

$$\int \frac{1}{1+e^x} dx$$

Add and subtract e^x in the numerator to manually split the fraction:

$$\int \frac{1+e^x - e^x}{1+e^x} dx = \int \left(1 - \frac{e^x}{1+e^x} \right) dx = x - \ln(1+e^x) + C$$

Example 7: Sneaky U-Substitution

$$\int \frac{\sec x \tan x}{\sec^2 x - \sec x} dx$$

Let $u = \sec x$, $du = \sec x \tan x dx$. The integral factors nicely:

$$\int \frac{1}{u^2 - u} du = \int \frac{1}{u(u-1)} du = \int \left(\frac{1}{u-1} - \frac{1}{u} \right) du = \ln|\sec x - 1| - \ln|\sec x| + C$$

7.3 Integration by Parts

Example 8

$$\int \frac{\cos(1/x)}{x^3} dx$$

Split $\frac{1}{x^3}$ into $\frac{1}{x} \cdot \frac{1}{x^2}$. Let $u = \frac{1}{x}$, $du = -\frac{1}{x^2} dx$, then use Integration by Parts:

$$-\int u \cos u du = -(u \sin u + \cos u) + C = -\frac{\sin(1/x)}{x} - \cos(1/x) + C$$

Example 9

$$\int x \sec x \tan x \, dx$$

Use integration by parts by picking $u = x$ and $dv = \sec x \tan x \, dx$:

$$x \sec x - \int \sec x \, dx = x \sec x - \ln |\sec x + \tan x| + C$$

Example 10

$$\int \ln(x^2 + 1) \, dx$$

Use integration by parts with $u = \ln(x^2 + 1)$ and $dv = dx$:

$$x \ln(x^2 + 1) - \int \frac{2x^2}{x^2 + 1} \, dx = x \ln(x^2 + 1) - 2 \int \left(1 - \frac{1}{x^2 + 1}\right) \, dx = x \ln(x^2 + 1) - 2x + 2 \arctan(x) + C$$

Example 11

$$\int \frac{\arctan(x)}{x^2} \, dx$$

Use integration by parts with $u = \arctan(x)$ and $dv = \frac{1}{x^2} \, dx$, then use partial fractions (or zero sub):

$$-\frac{\arctan(x)}{x} + \int \frac{1}{x(1+x^2)} \, dx = -\frac{\arctan(x)}{x} + \int \left(\frac{1}{x} - \frac{x}{1+x^2}\right) \, dx = -\frac{\arctan(x)}{x} + \ln|x| - \frac{1}{2} \ln(1+x^2) + C$$

7.4 Trigonometric Strategies and Symmetry**Example 12**

$$\int \frac{\sin^3 x}{\cos x} \, dx$$

Convert $\sin^2 x$ to $1 - \cos^2 x$, and substitute $u = \cos x$:

$$\int \frac{(1-u^2)(-du)}{u} = \int \left(u - \frac{1}{u}\right) \, du = \frac{\cos^2 x}{2} - \ln |\cos x| + C$$

Example 13

$$\int \frac{1 + \sin x}{1 + \cos x} \, dx$$

Split the fraction. Multiply the first part by the conjugate $1 - \cos x$, and use u -sub for the second:

$$\int \frac{1 - \cos x}{\sin^2 x} \, dx + \int \frac{\sin x}{1 + \cos x} \, dx = \int (\csc^2 x - \cot x \csc x) \, dx - \ln(1 + \cos x) = -\cot x + \csc x - \ln(1 + \cos x) + C$$

Example 14

$$\int \sqrt{\frac{1+x}{1-x}} \, dx$$

Multiply the numerator and denominator by the conjugate $\sqrt{1+x}$ to clear the root on top:

$$\int \frac{1+x}{\sqrt{1-x^2}} \, dx = \int \frac{1}{\sqrt{1-x^2}} \, dx + \int \frac{x}{\sqrt{1-x^2}} \, dx = \arcsin(x) - \sqrt{1-x^2} + C$$

Example 15: Symmetry Trap

$$\int_{-\pi/2}^{\pi/2} \frac{x}{1 + \cos x} dx$$

Always check symmetric bounds first! Since x is an odd function and $1 + \cos x$ is even, the overall product is odd:

$$\int_{-\pi/2}^{\pi/2} \frac{x}{1 + \cos x} dx = 0$$

7.5 Advanced and Forced U-Substitutions

Example 16

$$\int \arctan(\sqrt{x}) dx$$

Force substitution $u = \sqrt{x}$, yielding $dx = 2u du$, followed by integration by parts:

$$\int 2u \arctan(u) du = u^2 \arctan(u) - \int \frac{u^2}{1 + u^2} du = x \arctan(\sqrt{x}) - \sqrt{x} + \arctan(\sqrt{x}) + C$$

Example 17

$$\int (1 + \sqrt{x})^4 dx$$

Avoid expanding the 4th power mechanically! Force substitution $u = 1 + \sqrt{x}$, meaning $x = (u - 1)^2$ and $dx = 2(u - 1) du$:

$$\int u^4 \cdot 2(u - 1) du = 2 \int (u^5 - u^4) du = \frac{(1 + \sqrt{x})^6}{3} - \frac{2(1 + \sqrt{x})^5}{5} + C$$

Example 18: Hidden Fractional Power

$$\int \frac{\sqrt{x}}{1 + x^3} dx$$

Rewrite x^3 as $(x^{3/2})^2$. Let $u = x^{3/2}$, so $du = \frac{3}{2}\sqrt{x} dx$:

$$\frac{2}{3} \int \frac{1}{1 + u^2} du = \frac{2}{3} \arctan(x^{3/2}) + C$$

Example 19: The "Substitute Everything" Trick

$$\int \sqrt{1 + e^x} dx$$

Let u equal the entire root $u = \sqrt{1 + e^x}$, meaning $u^2 = 1 + e^x$ and $2u du = e^x dx = (u^2 - 1)dx$:

$$\int u \frac{2u}{u^2 - 1} du = 2 \int \left(1 + \frac{1}{u^2 - 1}\right) du = 2\sqrt{1 + e^x} - 2 \operatorname{artanh}(\sqrt{1 + e^x}) + C$$

Example 20

$$\int x^5 e^{-x^3} dx$$

Split x^5 into $x^3 \cdot x^2$. Let $u = x^3$, $du = 3x^2 dx$, then use integration by parts:

$$\frac{1}{3} \int u e^{-u} du = \frac{1}{3} (-u e^{-u} - e^{-u}) + C = -\frac{1}{3} e^{-x^3} (x^3 + 1) + C$$

Chapter 8

Integration Bee Training — Part 7.2: More Strategic Wildcards

This section continues the exploration of "wildcard" integrals from Chapter 7, focusing on speed integration strategies such as the caveman trick, advanced forced substitutions, algebraic/trigonometric conjugation, and symmetry traps.

8.1 Product Rule and Algebraic Factoring

Example 1: The Product Rule Trap

$$\int \frac{e^x(x-1)}{x^2} dx$$

Instead of using integration by parts, recognize this as a direct application of the quotient/product rule. Splitting the terms inside the integrand yields:

$$\int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$$

Since $\frac{d}{dx} \left(\frac{1}{x} \right) = -\frac{1}{x^2}$, this perfectly matches the expansion of $\frac{d}{dx} \left(\frac{e^x}{x} \right)$:

$$\int \frac{e^x(x-1)}{x^2} dx = \frac{e^x}{x} + C$$

Example 2: The Caveman Trick

$$\int \frac{1}{x(x^4+1)} dx$$

Avoid tedious partial fractions. Instead, force a u -substitution by factoring out the highest power (x^4) from the denominator:

$$\int \frac{1}{x \cdot x^4(1+x^{-4})} dx = \int \frac{x^{-5}}{1+x^{-4}} dx$$

Let $u = 1 + x^{-4}$, which means $du = -4x^{-5} dx$:

$$-\frac{1}{4} \int \frac{1}{u} du = -\frac{1}{4} \ln(1+x^{-4}) + C$$

Example 3: Power Expansion and Wallis' Formula

$$\int_0^{\pi/2} (x + \sin x)^2 dx$$

Completely expand the binomial expression before evaluating the individual pieces:

$$\int_0^{\pi/2} (x^2 + 2x \sin x + \sin^2 x) dx$$

Evaluating term-by-term:

1. $\int_0^{\pi/2} x^2 dx = \left[\frac{x^3}{3} \right]_0^{\pi/2} = \frac{\pi^3}{24}$
2. $2 \int_0^{\pi/2} x \sin x dx$ evaluated via integration by parts yields $2 [\sin x - x \cos x]_0^{\pi/2} = 2(1 - 0) = 2$
3. $\int_0^{\pi/2} \sin^2 x dx$ can be solved immediately using Wallis' Product rule: $\frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi}{4}$

Summing the terms together gives the final value:

$$\frac{\pi^3}{24} + 2 + \frac{\pi}{4}$$

Example 4

$$\int \frac{1}{\sqrt{x} + x\sqrt{x}} dx$$

Factor out $\sqrt{x}$ from the denominator to make the underlying structure clearer:

$$\int \frac{1}{\sqrt{x}(1+x)} dx$$

Let $u = \sqrt{x}$, meaning $du = \frac{1}{2\sqrt{x}} dx \implies 2 du = \frac{1}{\sqrt{x}} dx$:

$$\int \frac{2}{1+u^2} du = 2 \arctan(u) + C = 2 \arctan(\sqrt{x}) + C$$

8.2 Trigonometric Conversions and Forced Substitutions

Example 5

$$\int \frac{1}{1 + \cos^2 x} dx$$

There is no direct standard identity for $1 + \cos^2 x$. Multiply the numerator and denominator by $\sec^2 x$ to switch into a coordinate system driven by tangents and secants:

$$\int \frac{\sec^2 x}{\sec^2 x + 1} dx = \int \frac{\sec^2 x}{(\tan^2 x + 1) + 1} dx = \int \frac{\sec^2 x}{\tan^2 x + 2} dx$$

Let $u = \tan x$, meaning $du = \sec^2 x dx$:

$$\int \frac{1}{u^2 + 2} du = \frac{1}{\sqrt{2}} \arctan\left(\frac{u}{\sqrt{2}}\right) + C = \frac{1}{\sqrt{2}} \arctan\left(\frac{\tan x}{\sqrt{2}}\right) + C$$

Example 6

$$\int \sqrt{x} e^{\sqrt{x}} dx$$

Eliminate the radical fraction setting $u = \sqrt{x}$, meaning $u^2 = x$ and $2u du = dx$:

$$\int u \cdot e^u \cdot (2u du) = 2 \int u^2 e^u du$$

Applying tabular integration by parts on $2 \int u^2 e^u du$:

$$2e^u(u^2 - 2u + 2) + C = 2e^{\sqrt{x}}(x - 2\sqrt{x} + 2) + C$$

Example 7: The "Root Elimination" Strategy

$$\int \frac{1}{\sqrt{\sqrt{x} + 1}} dx$$

When dealing with deeply nested radicals, set u to equal the entire expression: $u = \sqrt{\sqrt{x} + 1}$. Squaring both sides yields $u^2 = \sqrt{x} + 1 \implies \sqrt{x} = u^2 - 1$. Squaring again gives $x = (u^2 - 1)^2$. Compute the differential dx :

$$dx = 2(u^2 - 1) \cdot 2u du = 4u(u^2 - 1) du$$

Substitute these back into the integral:

$$\int \frac{1}{u} \cdot 4u(u^2 - 1) du = 4 \int (u^2 - 1) du = 4 \left(\frac{u^3}{3} - u \right) + C$$

Substituting $u = \sqrt{\sqrt{x} + 1}$ back gives:

$$\frac{4}{3}(\sqrt{x} + 1)^{3/2} - 4\sqrt{\sqrt{x} + 1} + C$$

Example 8: Double Angle Chain Rule

$$\int \frac{\sin 2x}{1 + \sin^4 x} dx$$

Recall that the derivative of $\sin^2 x$ is $2 \sin x \cos x = \sin 2x$. Let $u = \sin^2 x$, which gives $du = \sin 2x dx$:

$$\int \frac{1}{1 + u^2} du = \arctan(u) + C = \arctan(\sin^2 x) + C$$

Example 9: Secret Logarithmic Substitution

$$\int \frac{\ln(\tan x)}{\sin x \cos x} dx$$

Test the derivative of the target function. Let $u = \ln(\tan x)$:

$$du = \frac{1}{\tan x} \cdot \sec^2 x dx = \frac{\cos x}{\sin x} \cdot \frac{1}{\cos^2 x} dx = \frac{1}{\sin x \cos x} dx$$

This perfectly eliminates the denominator, turning the problem into a fundamental power-rule integral:

$$\int u du = \frac{1}{2}u^2 + C = \frac{1}{2}(\ln(\tan x))^2 + C$$

8.3 Conjugation and Advanced Definite Integrals

Example 10

$$\int \frac{1}{\sqrt{x+1} + \sqrt{x}} dx$$

Multiply the numerator and denominator by the conjugate expression $\sqrt{x+1} - \sqrt{x}$:

$$\int \frac{\sqrt{x+1} - \sqrt{x}}{(x+1) - x} dx = \int (\sqrt{x+1} - \sqrt{x}) dx$$

Apply the power rule directly:

$$\frac{2}{3}(x+1)^{3/2} - \frac{2}{3}x^{3/2} + C$$

Example 11

$$\int \frac{\ln(x+1)}{x^2} dx$$

Apply integration by parts with $u = \ln(x+1)$ and $dv = \frac{1}{x^2} dx \implies du = \frac{1}{x+1} dx$ and $v = -\frac{1}{x}$:

$$-\frac{\ln(x+1)}{x} + \int \frac{1}{x(x+1)} dx$$

Using a zero-substitution $(+x - x)$ inside the remaining integral yields $\int \left(\frac{1}{x} - \frac{1}{x+1}\right) dx$:

$$-\frac{\ln(x+1)}{x} + \ln|x| - \ln|x+1| + C$$

Example 12

$$\int 2^x \cdot 5^x dx$$

Consolidate the base terms utilizing standard exponent algebra rules:

$$\int (2 \cdot 5)^x dx = \int 10^x dx = \frac{10^x}{\ln 10} + C$$

Example 13

$$\int \frac{1}{x \ln x - x} dx$$

Factor out x from the denominator to establish a clear u -substitution pathway:

$$\int \frac{1}{x(\ln x - 1)} dx$$

Let $u = \ln x - 1 \implies du = \frac{1}{x} dx$:

$$\int \frac{1}{u} du = \ln|u| + C = \ln|\ln x - 1| + C$$

Example 14

$$\int \sqrt{1 - \sin x} \, dx$$

Multiply the expression inside the radical by the trigonometric conjugate $\frac{1+\sin x}{1+\sin x}$:

$$\int \sqrt{\frac{1 - \sin^2 x}{1 + \sin x}} \, dx = \int \frac{\cos x}{\sqrt{1 + \sin x}} \, dx$$

Let $u = 1 + \sin x \implies du = \cos x \, dx$:

$$\int \frac{1}{\sqrt{u}} \, du = 2\sqrt{u} + C = 2\sqrt{1 + \sin x} + C$$

Example 15

$$\int \frac{x^2 + 2}{x + 2} \, dx$$

Perform a nested zero-substitution to perfectly divide the polynomial without long division. Add and subtract $2x$ then add and subtract 4:

$$\int \frac{x^2 + 2x - 2x - 4 + 6}{x + 2} \, dx = \int \frac{x(x + 2) - 2(x + 2) + 6}{x + 2} \, dx = \int \left(x - 2 + \frac{6}{x + 2} \right) \, dx$$

Integrating the simplified polynomial terms:

$$\frac{x^2}{2} - 2x + 6 \ln |x + 2| + C$$

Example 16: Rationalizing Fractional Powers

$$\int_{-1}^0 \frac{\sqrt[3]{x} + 1}{\sqrt[3]{x} - 1} \, dx$$

Let $u = \sqrt[3]{x} \implies u^3 = x \implies dx = 3u^2 \, du$. Update the bounds: $x = -1 \implies u = -1$ and $x = 0 \implies u = 0$:

$$\int_{-1}^0 \frac{u + 1}{u - 1} \cdot 3u^2 \, du = 3 \int_{-1}^0 \frac{u^3 + u^2}{u - 1} \, du$$

Using polynomial division, rewrite the integrand:

$$3 \int_{-1}^0 \left(u^2 + 2u + 2 + \frac{2}{u - 1} \right) \, du = 3 \left[\frac{u^3}{3} + u^2 + 2u + 2 \ln |u - 1| \right]_{-1}^0$$

Evaluating at the upper and lower limits results in the final scalar value:

$$4 - 6 \ln 2$$

Example 17: Symmetry Trap with Absolute Value

$$\int_{-3}^3 \frac{x|x|}{x^2 + 1} \, dx$$

Before performing any algebraic steps, analyze the bounds $[-3, 3]$ for symmetry. The absolute value function $|x|$ is even, and $x^2 + 1$ is even, making their combination even. However, the

isolated x in the numerator is an odd function. An odd function multiplied by an even function creates an overall odd integrand:

$$f(-x) = \frac{-x|x| - x|}{(-x)^2 + 1} = -\frac{x|x|}{x^2 + 1} = -f(x)$$

Because the integrand is an odd function integrated over symmetric bounds, it evaluates directly to zero:

$$\int_{-3}^3 \frac{x|x|}{x^2 + 1} dx = 0$$

Example 18: Advanced Rational Exponential Substitution

$$\int_0^{\ln 10} \frac{e^x \sqrt{e^x - 1}}{e^x + 8} dx$$

Let $u = \sqrt{e^x - 1} \implies u^2 = e^x - 1 \implies 2u du = e^x dx$. Note that $e^x = u^2 + 1$, so the denominator becomes $(u^2 + 1) + 8 = u^2 + 9$. Convert the boundary conditions: $x = 0 \implies u = 0$ and $x = \ln 10 \implies u = \sqrt{10 - 1} = 3$:

$$\int_0^3 \frac{u \cdot (2u du)}{u^2 + 9} = 2 \int_0^3 \frac{u^2}{u^2 + 9} du$$

Apply a zero-substitution $(+9 - 9)$ inside the numerator:

$$2 \int_0^3 \left(1 - \frac{9}{u^2 + 9} \right) du = 2 \left[u - 3 \arctan \left(\frac{u}{3} \right) \right]_0^3$$

Substitute the boundaries:

$$2(3 - 3 \arctan(1)) - 0 = 2 \left(3 - \frac{3\pi}{4} \right) = 6 - \frac{3\pi}{2}$$

Example 19: Radical Fractional Grouping

$$\int \frac{1}{\sqrt{x + x^{3/2}}} dx$$

Factor out x inside the square root to parse the functions cleanly:

$$\int \frac{1}{\sqrt{x(1 + \sqrt{x})}} dx = \int \frac{1}{\sqrt{x} \sqrt{1 + \sqrt{x}}} dx$$

Let $u = 1 + \sqrt{x} \implies du = \frac{1}{2\sqrt{x}} dx \implies 2 du = \frac{1}{\sqrt{x}} dx$:

$$\int \frac{2}{\sqrt{u}} du = 4\sqrt{u} + C = 4\sqrt{1 + \sqrt{x}} + C$$

Example 20: Tangent Fraction Component Splitting

$$\int \frac{1 - \tan x}{1 + \tan x} dx$$

Convert the tangents into fundamental sine and cosine functions:

$$\int \frac{1 - \frac{\sin x}{\cos x}}{1 + \frac{\sin x}{\cos x}} dx = \int \frac{\cos x - \sin x}{\cos x + \sin x} dx$$

Let $u = \cos x + \sin x$. The differential is $du = (-\sin x + \cos x) dx$, which perfectly matches the numerator:

$$\int \frac{1}{u} du = \ln |u| + C = \ln |\cos x + \sin x| + C$$